

PRAJWAL SINGH

+91-9302967344 | prajwalsingh344@gmail.com | [LinkedIn](#) | [GitHub](#) | Bhopal, MP

PROFESSIONAL SUMMARY

B.Tech Computer Science student (2026) with hands-on experience in Java, C++, Python, PostgreSQL, and MySQL. Strong foundation in data structures, algorithms, object-oriented programming, and software development. Built backend applications and machine learning projects using Java, JDBC, PostgreSQL, Python, and scikit-learn, with experience in database design, integration testing, debugging, and Git-based development workflows. Passionate about developing reliable software systems and applying engineering principles to solve complex technical problems.

TECHNICAL SKILLS

Languages: Java, C++, Python, JavaScript, HTML, CSS

Java & Backend: OOP, Collections Framework, JDBC, Exception Handling, CRUD Applications, RESTful APIs, JSON, API Integration

Dev Tools: IntelliJ IDEA, VS Code, Microsoft Visual Studio, Git, GitHub, PyCharm, Jupyter Notebook

Databases: PostgreSQL, MySQL (Schema Design, Normalisation, Indexing, Query Optimisation)

CS Concepts: Data Structures & Algorithms, OOP, Modular Design, SDLC, Computer Architecture, RESTful Web Services

OS & Other: Windows 10/11, Linux (basic), Secure Coding Awareness

PROJECTS

Student Management System | *Java, PostgreSQL, JDBC, REST concepts, IntelliJ IDEA* | [GitHub](#)

- Developed a backend Java application with layered modular architecture, implementing full CRUD operations via JDBC with transaction management, input validation, and exception handling to ensure data integrity.
- Designed a PostgreSQL relational schema with constraints, referential integrity, and optimised queries; applied OOP principles — encapsulation, abstraction, and modular design — for testability and maintainability.
- Independently debugged and resolved data-layer issues, practising clean, readable code aligned with professional software development standards. Managed source code using Git and GitHub with regular commits and version control.

Alumni Management Platform | *Java, MySQL, Git* | [GitHub](#)

- Collaborated in a team of 4 using Git and GitHub workflows, owning the full database layer and participating in code reviews — mirroring Agile team practices.
- Designed a normalised MySQL schema with indexing for faster query performance across search, directory, and filtering features.
- Performed integration testing across search, filtering, and directory modules — validating data consistency and system behaviour end-to-end.

Movie Recommendation System | *Python, scikit-learn, Pandas, Jupyter* | [GitHub](#)

- Built a content-based recommendation engine using cosine similarity to return the top 5 similar movies from a 5,000+ record dataset using Python, Pandas, and scikit-learn.
- Independently learned and applied scikit-learn, demonstrating ability to quickly ramp up on new libraries and tools.
- Applied ML techniques (cosine similarity, vector operations) to a real-world dataset — demonstrating foundational AI/ML thinking and ability to prototype solutions quickly.

EDUCATION

Patel College of Science and Technology

B.Tech – Computer Science and Engineering

Bhopal, MP

2022 – 2026

Pooja Convent

Class XII (2022) | Class X (2020)

Bhopal, MP

2020 – 2022

CERTIFICATIONS

Core Java Certificate – Infosys Springboard

OOP, collections framework, JDBC, exception handling

May 2025

Python Developer Certificate – SoloLearn

Python fundamentals, file handling, OOP concepts

Mar 2025

KEY STRENGTHS

Strong Java, C++ & OOP fundamentals · Database design & query optimisation · Quick learner of new tools and environments · Collaborative team player · Detail-oriented with clean, maintainable code practices · Proficient in multiple programming languages · Strong system integration and testing mindset